

Adrian A. Charles

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EDUCATION

Queen's University

Kingston, Ontario

B.A.S in Computer Engineering

May 2026

- **Related Coursework:** Data Structures & Algorithms, Computer Organization & Programming, Object-Oriented Programming, Logic for Computer Science, Machine Learning and Deep Learning, Database Management Systems

EXPERIENCE

GE Vernova

Peterborough, Ontario

Engineering Intern

May 2024 – Aug 2024

- Designed and Developed a report generation tool using PHP, reducing manual engineering report generation and improving report data storage.
- Built and deployed a web UI (HTML, CSS) to easily allow engineers to generate standardized reports on demand.
- Maintained and fixed existing internal tools by adding new feature upgrades, improving reliability and debugging issues/bugs. Demonstrating ownership of deployed systems.
- Collaborated with other engineering interns and stakeholders to deliver data driven tools based on operational requirements.

GE Vernova

Peterborough, Ontario

Engineering Intern

May 2023 – Aug 2023

- Developed full-stack internal applications (PHP, MySQL, JavaScript) including a Customer Service Dashboard and Document Locator Tool to improve data accessibility and workflow efficiency.
- Designed and queried relational databases using SQL, for use in a structured data retrieval tool.
- Worked alongside engineers and interns in a large team environment, gaining exposure to mechanical engineering practices.

Queen's University - Faculty of Engineering and Applied Science

Kingston, Ontario

Teaching Assistant - Intro to Computer Programming

Sept 2023 – Dec 2023

- Supported first year engineering students in C in programming, debugging, and algorithmic problem solving.
- Graded assignments and provided viable technical feedback, strengthening knowledge of complex C and programming concepts to non-technical and beginner students.
- Balanced academic and instructional responsibilities, demonstrating time management and delivery under deadline.

PROJECTS

AWS Cloud Engineer Agent – AI Cloud Infrastructure Assistant

Kingston, Ontario

Developer

February 2026

- Developed an intelligent AWS Cloud Engineer agent using Python, Strands Agents SDK, and Amazon Bedrock, enabling natural language interaction for inspecting and reasoning about AWS infrastructure.
- Integrated AWS APIs via boto3 to retrieve and analyze information from services such as EC2, S3, CloudWatch, and IAM, supporting automated cloud infrastructure inspection and operational insights.
- Built a Streamlit-based conversational interface and incorporated external tools for AWS documentation retrieval and architecture diagram generation, allowing users to explore cloud systems and visualize infrastructure through natural language queries.

NBA Predictive Analytics Pipeline – Data Engineering + Machine Learning Project

Kingston, Ontario

Developer

November 2025

- Designed and implemented an end to end ETL pipeline, which ingests, cleans, transforms and stores NBA game log data using Python, SQLite, and external REST API's
- Engineered structured data models and features like lagged performance windows, opponent metrics, and usage indicators to support analytics and ML workloads.
- Built and evaluated a RandomForestRegressor model, achieving strong predictive performance on unseen game data through systematic feature engineering.
- Developed an analytics layer incorporating data validation, anomaly detection, error analysis, and visualization to ensure data quality and model explainability.
- Applied production style data engineering patterns, including version control using GitHub.

SKILLS

Programming: Python, SQL, Java, C, JavaScript, PHP, MATLAB, Pandas, SQLite, MySQL

Tools: AWS, Databricks, GitHub, VS Code, PyCharm, Eclipse